C1DB1B: Datahantering för beslutsfattande

Document B: The Data Stream

Dataströmmen

Versioned datasets, dictionaries, and module releases

Course Team

Current offering | Vintage as stated below

Why this document exists

The assignment (`project_Introduction.tex`) and the module handbook (`module_design.tex`) describe *what students decide* and *how they work*. They should not contain a particular year's population counts, clinic lists, or SCB table vintages. This document is the data contract for **course-issued** files. When Statistics Sweden, VGR, or the course team updates those files, only the folders under `data/00`–`data/04` and `manifest.json` change.
Notebooks *read* the manifest. Students do not edit it.
Group-collected downloads (VGR, SCB, or other public extracts for the group's approved KPIs) are the expansion of that architecture. They follow a different path, described below. They are not optional extras once Module 1 has Verified them.

1 How to Change Data Over Time

The course uses one folder tree:

```
data/
  manifest.json
  00_decision_model_zero/
  01_module1/
  02_module2/
  03_module3/
  04_module4/
```

`data/manifest.json` is the single switchboard. It records:

- the course vintage (for example `2026.1`);
- which file each module must load;
- the reference years t_0 and t_1 used in calculations;
- the official source table identifiers and retrieval dates.

To refresh a module's data without rewriting teaching materials:

1. Place the new extract in the relevant module folder. Keep the previous file if you need an audit trail (rename with a date suffix).
2. Update the corresponding paths, years, and source notes in `manifest.json`.
3. Update the vintage line at the top of this document and the module source note.
4. Re-run the instructor solution notebook for that module and store new reference values if the live challenge depends on them.

Notebooks must load years and paths from the manifest:

```
MANIFEST = json.loads(Path("data/manifest.json").read_text(encoding="utf-8"))
t0 = MANIFEST["module1"]["t0"]
t1 = MANIFEST["module1"]["t1"]
anchor = Path(MANIFEST["module1"]["anchor_file"])
bench = Path(MANIFEST["module3"]["benchmark_file"])
```

Do not write `2015` or `2025` as literals in student tasks unless you are illustrating a historical example.

The numbered steps above are for the **course team** when a new vintage is issued. They are not student tasks.

2 Course Files versus Group-Collected Files

Students work with two layers of data.

Layer	Who writes the files	How notebooks find them
Course-issued (anchor, companion, benchmark, scenarios)	The course team.	Read paths and years from `data/manifest.json`. Do not edit that file.
Group-collected (extra VGR, SCB, or other public extracts for approved KPIs)	The group.	Save a CSV under `data/group_sources/` and load it with an explicit path in the notebook. Record it in the group's source register, not in `manifest.json`.

2.1 What students do with a VGR download

Yes: they format it so it can join the five-municipality table. No: they do not wire it through `manifest.json`.

1. Keep the original download (for example a PDF table or Excel export) outside the issued folders, or in `data/group_sources/raw/`.
2. Do not overwrite the downloaded file. In the notebook, standardize keys and rename columns to the field names below (`Municipality_Code`, `Year`, ...).
3. Save one analysis-ready CSV per source, for example `data/group_sources/travel_times.csv`.
4. Add one row to the group's source register (`data/group_sources/Source_Register.csv`): file path, observation unit, keys, coverage, units, missing-value markers.
5. Join with the same Module 2 rules: `how`, `validate`, `indicator`. Keep all five municipalities. A missing match is a coverage gap, not a zero.

A CSV is enough. Students are not asked to write JSON. A broken comma in `manifest.json` would stop the whole class pipeline; a broken group CSV affects only that group and is visible in the source register.

If a group cannot produce a comparable extract for all five municipalities, the indicator stays Unresolved or Unknown. That is an acceptable Module 1–2 outcome.

There is no advanced, source-specific support for group downloads. The course team does not know in advance which table a group will find. Formatting, name-to-code mapping, and the join are the group's work, using the issued companions as the method example.

Do not put group files here

`data/00_decision_model_zero/` through `data/04_module4/` and `data/manifest.json` are course-issued. Replacing them mixes vintages and breaks the live challenge.

Current vintage

Vintage: 2026.1
Released: with this vintage (not a timetable date)
Anchor reference years: $t_0 = 2015$, $t_1 = 2025$
Geographic coverage: 52 municipalities present in the SCB Västra Götaland extract (including three Halland-border codes used in the source table: 1382 Falkenberg, 1383 Varberg, 1384 Kungsbacka).
Predictive component: not issued as a common file. Optional for a group: own model (Module 2 may hold out about 20% of eligible observations) or an external forecast, either way justified. Otherwise mark Not assigned.
Module 3 common exercise: synthetic WSM/TOPSIS benchmark, issued.
Module 4 common exercise: scenario specification and teaching cost ceiling, issued. Not a predictive hold-out.

3 Document Map

Document	Role	Changes when?
`project_Introduction.tex`	The Vårdlyftet assignment	Only if the decision question changes.
`module_design.tex`	Module tasks, gateways, methods	Only if pedagogy or assessment changes.
`lectures2026/`	Preparation lectures	When teaching sequence or examples change.
`Data_Stream.tex` (this file)	Data contract, dictionaries, vintages	When sources, years, or files change.
`data/manifest.json`	Machine-readable switchboard	Every data release.

4 Release 00: Decision Model Zero

Guiding question from the assignment: Which three portfolios would the group prefer to work with, before systematic methods are taught?

Files

- `data/00_decision_model_zero/VGRBiddingDataset.csv`
- Optional Excel copy: `data/00_decision_model_zero/VGRBiddingDataset.xlsx`

What students see

Municipalities appear as coded keys (`MUN-VGR-101`, ...). Each row is one municipality inside a five-municipality portfolio (`P1`–`P7`).

Fields

Field	Type	Meaning
`Portfolio ID`	code	Group of five coded municipalities.
`Anonymized Key`	code	Municipality disguise before allocation.
`VGR Region Context`	text	Instructor-only identity (strip before student release if required).
`Total Population (2024/2025)`	integer	Snapshot population.
`Elderly Share % (65+)`	percent	Broad elderly share, not the Module 1 80+ indicator.
`Operating Clinics`	integer	Approximate incumbent count.
`Avg Household Income (SEK/mo)`	SEK	Snapshot income.
`Est. CNI Score`	index	Care Need Index relative to 1.0 national average.
`Primary Trade-off Tension`	text	Qualitative tension, not a scored KPI.

Release rule

If identities must remain hidden until after allocation, publish a student copy with `VGR Region Context` removed. Keep the full file in the instructor folder.

After allocation, students work with official SCB municipality codes from Release 01. Do not treat Decision Model Zero figures as the Module 1 calculation source.

5 Release 01: Module 1 Anchor

Guiding question from the handbook: What should matter in this decision, and can the group calculate a common demographic baseline for its five municipalities?

The course team provides one regional working table. All groups use it. Each group then selects the five allocated municipality codes.

Files

- `data/01_module1/VGR_Demographics_Anchor.csv`
- `data/01_module1/SOURCE_NOTE.md`
- Path and years in `data/manifest.json` under `module1`

Grain

One row = one municipality × one calendar year.

Required fields

Field	Type	Definition
`Municipality_Code`	identifier	Official SCB kommun-kod. Not a measurement.
`Municipality_Name`	text	Official name in the extract.
`Year`	integer	Calendar year of the stock count.
`Pop_Total`	count	Total population, all ages.
`Pop_0_17`	count	Ages 0–17.
`Pop_18_64`	count	Ages 18–64.
`Pop_65_79`	count	Ages 65–79.
`Pop_80_Plus`	count	Ages 80 and above, including 100+.

Provenance (vintage 2026.1)

- **Producer:** Statistics Sweden (SCB), table BE0101, population by region, age, sex, and year.
- **Underlying extract:** `VGR_Demographics_2015_2025.csv`, itself curated from `VGR_Demographics_Raw.xlsx`.
- **Construction:** Gender `totalt` only; age `TOT` excluded; single-year ages aggregated into the four bands above; county total (`region_code = 14`) excluded.
- **Integrity rule:** `Pop_0_17` + `Pop_18_64` + `Pop_65_79` + `Pop_80_Plus` = `Pop_Total`.

Prescribed indicators (formulas stay in the handbook)

Notebooks must take t_0 and t_1 from the manifest.

$$S_{80+,t_1} = 100\,\frac{P_{80+,t_1}}{P_{\text{total},t_1}},$$

$$D_{t_1} = 100\,\frac{P_{0\text{–}17,t_1} + P_{65\text{–}79,t_1} + P_{80+,t_1}}{P_{18\text{–}64,t_1}},$$

$$G_{80+} = 100\,\frac{P_{80+,t_1} - P_{80+,t_0}}{P_{80+,t_0}}.$$

Do not compute mean or median age from these four bands.

Structural checks the notebook must run

1. All five allocated municipality codes are present.
2. Each municipality has one row for t_0 and one row for t_1.
3. `Municipality_Code` and `Year` jointly identify rows uniquely.
4. Required population fields are numeric, non-negative, and not missing.
5. The four age-group counts sum to `Pop_Total`.

If a check fails, students inspect extraction or selection. They must not drop a municipality, replace missing values with zero, or edit the source file.

Refreshing Module 1

When SCB publishes a later year, rebuild `VGR_Demographics_Anchor.csv`, move t_1 forward in the manifest, and keep t_0 either fixed (ten-year window) or shifted (rolling window). Record the choice in the source note.

6 Release 02: Module 2 Companion Streams

Guiding question from the handbook: Can the group combine the sources needed to compare its five municipalities without losing alternatives, inventing values, or silently changing indicator meaning?

Module 2 consumes:

- the frozen Module 1 anchor for the five allocated municipalities (Release 01);
- the common companion files in this release;
- any additional verified sources from the group's approved KPI-to-data table.

Complete the common companion exercise first. Then apply the same checks to portfolio sources. There is no fixed dataset count.

Files

- `data/02_module2/VGR_Companion_Annual.csv`
- `data/02_module2/VGR_Companion_StaticLookup.csv`
- `data/02_module2/VGR_Masked_TreatmentExercise.csv`
- `data/02_module2/SOURCE_NOTE.md`
- Paths and status in `data/manifest.json` under `module2`

Predictive component (vintage 2026.1)

Not issued as a common file. Prediction is optional for a group: build a model (Module 2 may hold out about 20% of eligible observations) or join a published forecast, and justify either. Do not split the five-municipality decision table as an MCDA training/test set. Do not split the ten-row Module 1 prototype to create a hold-out file. If the group does not take it up, mark `Not assigned`.

A. Annual companion

Grain: one row = one municipality in year t_1 (the year is stored in the file and must match `module1.t1` in the manifest).

Join: `Municipality_Code` (and `Year`, after confirming both tables use the same t_1). Expected relationship to the t_1 demographic extract: one-to-one.

Field	Type	Definition
`Municipality_Code`	identifier	Official SCB kommun-kod.
`Municipality_Name`	text	Display label only.
`Year`	integer	Calendar year of the observation (t_1).
`Operating_Clinics`	count	Course-constructed incumbent count. Not a live VGR clinic register.
`Clinics_per_10k`	rate	$10000 \times$ `Operating_Clinics`/`Pop_Total` at t_1. Verify; do not treat as an independent measurement.
`Avg_Travel_Time_min`	minutes	Curated accessibility estimate. Documented missing marker: `...`

Documented travel-time gaps in vintage 2026.1: `1444` Grästorp, `1446` Karlsborg, `1447` Gullspång, `1465` Svenljunga.

B. Static municipality lookup

Grain: one row = one municipality. There is **no** `Year` field.

Join: `Municipality_Code` only. After the annual table has been reduced to one row per municipality, the expected relationship is one-to-one. If annual history is joined before filtering to t_1, the relationship is many-to-one.

Do not invent a year column on this file. If a static value is reused at t_1, label the reference period and the assumption.

Field	Type	Definition
`Municipality_Code`	identifier	Official SCB kommun-kod.
`Municipality_Name`	text	Display label only.
`Care_Need_Index`	index	Course-constructed CNI relative to 1.0. Teaching values aligned with Decision Model Zero where identities match; otherwise 1.0.
`Est_Premises_Cost_kSEK`	kSEK	Course-constructed premises-cost teaching estimate. Documented missing marker: `..` (vintage 2026.1: `1438` Dals-Ed).

C. Masked treatment exercise

Grain: one synthetic record. These rows are **not** municipalities and must not enter the decision table.

Field	Type	Definition
`Record_ID`	identifier	Teaching record label.
`Synthetic_Value`	numeric or `..`	Observed values 10, 12, 14, 64 and one documented missing marker.
`Note`	text	`observed`, `observed outlier`, or `documented missing`.

Use this file for the required two-treatment comparison when the portfolio has no suitable missingness issue, or as the common mean-versus-median rehearsal. Keep results separate from factual establishment evidence.

Missing-value rule

The documented marker in this release is `...` Parse it as missing. It is not zero, not "no clinics", and not "zero minutes".

Do not treat every blank, dash, or unmatched join as the same event:

- missing in the source;
- unmatched after a join;
- not applicable or suppressed, where documented;
- unavailable because coverage differs.

Required pipeline checks

1. Every input has an observation unit, key fields, coverage, units, and missing-value markers in the source register.
2. Keys are standardized in code. Source files remain unmodified.
3. Key uniqueness is checked before each join.

4. Joins declare `how`, `validate`, and `indicator`.
5. The audit records input counts, output count, unmatched keys, and the join relationship.
6. All five allocated codes remain in the decision table.
7. A missing match is reported, not filled with zero and not used as a reason to drop a municipality.
8. Predictive partitions are omitted unless the group took up a model in Module 2 (then hold out about 20% of eligible observations). An external forecast is joined like any other group source. Do not split the five municipalities.

What these files are not

They are not a live clinic register, a cadastral premises-cost extract, or an official Care Need Index publication. Labelled teaching fields must not be presented as independently verified establishment facts. Group-collected external sources remain the group's responsibility and must pass the same checks.

Refreshing Module 2

Replace files only inside `data/02_module2/`. Update `manifest.json` paths, the predictive-component flag, and this section. Re-execute the instructor solution notebook. If a later vintage assigns prediction, publish the modelling specification before requiring a split.

7 Release 03: Module 3 Benchmark

Guiding question from the handbook: Under explicit preferences, which alternative performs best in a checked comparison, and what would change that conclusion?

Module 3 consumes:

- the group's approved five-municipality decision table (produced in Module 2 from Releases 01 and 02);
- the common synthetic benchmark in this release;
- the group's recorded indicator directions and baseline weights.

The course team does **not** issue a new municipal extract for Module 3. The benchmark is a calculation exercise. Portfolio scores are computed from the Module 2 table.

Complete and check the benchmark in `starter_m3.ipynb` before applying the same functions to the portfolio.

Files

- `data/03_module3/MCDA_Benchmark.csv`
- `data/03_module3/MCDA_Benchmark_Meta.csv`
- `data/03_module3/SOURCE_NOTE.md`
- Paths and status in `data/manifest.json` under `module3`

Predictive component (vintage 2026.1)

Not issued as a common file. Insert a predicted column only if the group justified it in Module 2 and the value is finite for all five municipalities. Label it as a prediction. Otherwise mark `Not assigned`. Do not invent a regression target merely to fill the matrix.

A. Benchmark matrix

Grain: one row = one synthetic alternative (`A`–`E`). These labels are not municipalities and must not enter the portfolio decision table.

Field	Type	Definition
`Alternative`	identifier	Teaching label A–E.
`opportunity`	index	Benefit criterion; higher preferred.
`annual_cost`	SEK million	Cost criterion; lower preferred.

Values (vintage 2026.1):

Alternative	opportunity	annual_cost
A	10	10
B	8	6
C	6	4
D	4	3
E	2	2

B. Benchmark metadata

Grain: one row = one scored indicator.

Field	Type	Definition
`indicator`	name	Must match a column in the benchmark matrix. Align weights and directions by this name, not by column order.
`direction`	`Benefit` or `Cost`	Benefit: higher preferred. Cost: lower preferred.
`weight`	$[0, 1]$	Non-negative. The issued baseline sums to 1: opportunity 0.60, annual cost 0.40.

Method conventions (lecture, notebook, and this contract)

- WSM uses direction-aware min–max normalization. A constant column is scored 0 for every alternative; retain its weight.
- TOPSIS vector-normalizes the **raw** matrix, then applies weights. Do not feed the WSM Z matrix into TOPSIS.
- Cost criteria keep non-negative weights. Direction is handled in normalization, not by a negative w.
- Rank higher scores first. Exact ties use average rank.
- Spearman ρ is descriptive. It is undefined if a ranking is constant. No significance test is required for five alternatives.

Reference checks (vintage 2026.1)

WSM scores: A 0.60, B 0.65, C 0.60, D 0.50, E 0.40.

TOPSIS closeness (three decimal places): A 0.565, B 0.648, C 0.588, D 0.487, E 0.435.

Live-challenge rehearsal on this same file: set annual-cost weight to 0.70 and redistribute. WSM then ranks E first with score 0.70. The in-class instruction may use a different α; it still uses this benchmark, not an unavailable portfolio indicator.

Portfolio scoring

A Module 2 column may enter the scoring matrix only if:

- it is an approved scored indicator;
- every one of the five municipalities has a finite value;
- Benefit/Cost direction is recorded;
- it is not an identifier, year, or status flag.

Exclude a source-missing field (vintage 2026.1 example: travel time for `1465` Svenljunga) unless an instructor-approved treatment exists. Do not fill with zero to force a rank. Do not drop a municipality to make a method run.

Clinic counts, CNI, and premises costs remain labelled teaching values. They may be scored only if they were approved in Module 1, and the label stays in the metadata.

What these files are not

They are not Västra Götaland municipalities, not a live cost extract, and not establishment evidence. Benchmark leaders are teaching results.

Refreshing Module 3

Replace files only inside `data/03_module3/`. Update `manifest.json` paths and the predictive-component flag. Re-execute `solution_m3.ipynb` and replace the reference scores in this section and in the source note if the benchmark values change.

8 Release 04

`data/04_module4/` issues the Module 4 scenario contract. It is not a predictive hold-out.

8.1 Files

File	Role
`Scenario_Spec.csv`	Named changes for baseline, operating, and combined scenarios on the synthetic benchmark and the portfolio cost field.
`Constraint_Spec.csv`	Teaching ceilings. The benchmark ceiling is compulsory for the common rehearsal. The portfolio ceiling is an illustrative demonstration assumption.
`SOURCE_NOTE.md`	Source note and live-challenge boundary.

8.2 Issued parameters

- Operating scenario: multiply the named cost field by 1.20. Keep other indicators and baseline weights unchanged.
- Combined scenario (benchmark): the same cost scaling and cost weight 0.70 (opportunity 0.30).
- Combined scenario (portfolio): the same cost scaling and the Module 3 premises-cost priority weight 0.40.
- Benchmark ceiling: `annual_cost` ≤ 6.5 cost units. Synthetic teaching assumption. Alternative B (6.0) meets it at baseline and exceeds it after the 20% increase (7.2). Alternative A (10) already exceeds it at baseline.

- Demonstration portfolio ceiling: `Est_Premises_Cost_kSEK` ≤ 650. Illustrative teaching assumption for instructor notebooks. Groups must label any portfolio ceiling as a group assumption unless a case requirement is issued.

8.3 What this release is not

Do not split the five-municipality table. Do not treat these files as reserved outcomes. Prediction remains **Not assigned** in vintage 2026.1.

Uniform positive scaling of one criterion leaves prescribed normalized values unchanged when normalization is recomputed. Absolute costs still rise. Recheck the ceiling.

To refresh this release, replace files only inside `data/04_module4/`, update `manifest.json`, re-execute `solution_m4.ipynb`, and revise the reference notes in this section if the issued multiplier or ceiling changes.

9 Field Naming Convention

Use the following names in every new extract so notebooks remain stable:

- `Municipality_Code` — SCB kommun-kod, stored as text or integer identifier, never used as a numeric feature.
- `Municipality_Name` — display label only.
- `Year` — calendar year of the observation.
- `Alternative` — synthetic benchmark row label in Module 3 only.
- `direction` — `Benefit` or `Cost`.
- `weight` — non-negative preference weight.
- Indicator-specific fields documented in the module source note, not invented ad hoc in notebooks.

If an upstream table uses Swedish headers, map them in the source note. Do not require students to guess column names from a live website.

10 Instructor Checklist for a Data Refresh

1. Confirm the assignment question in `project_Introduction.tex` is unchanged. If the decision changed, that is a course change, not a data change.
2. Replace files only inside `data/<module>/`.
3. Update `manifest.json` vintage, dates, paths, and t_0/t_1.
4. Update this document's vintage box and the relevant source note.
5. Execute the instructor solution notebook (`solution_m1.ipynb`, `solution_m2.ipynb`, `solution_m3.ipynb`, or `solution_m4.ipynb`) and archive reference outputs.
6. Tell students which vintage they must download. Mixed vintages in one group are not permitted.

The data stream in one sentence

Keep the Vårdlyftet decision fixed, keep the methods in the handbook fixed, and let `data/` plus this document absorb every change in years, tables, and extracts.